Linear Algebra and Its Applications

Section 2.S: Supplementary Exercises

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.S Chapter 2 supplementary exercises

These exercises combine matrix shapes, multiplication order, inverse identities, and the geometry of column and null spaces. Unless the statement explicitly says a matrix is square, do not add that hypothesis silently. A rectangular matrix can have a left inverse or a right inverse without having both. For a true–false question, a false statement is settled by one valid counterexample, while a true statement needs a general argument.

Exercise 2.S.1: Sixteen structural true–false statements

(a) **True.** If A, B are $m \times n$, then AB^T has shape $(m \times n)(n \times m)$ and $A^T B$ has shape $(n \times m)(m \times n)$. Both inner dimensions agree.

(b) **False.** The product has as many columns as B, not A. For example a 1×3 matrix times a 3×2 matrix has two columns, even though the first factor has three columns.

(c) **True.** For $A = \operatorname{diag}(a_1, \ldots, a_m)$, entry (i, j) of AB is $a_i b_{ij}$. Thus row i is scaled by a_i.

(d) **False.** Let $B = 0$, $C = I$, $D = 0$ of the same square size. Then $BC = BD = 0$ but $C \neq D$. Cancellation needs an injective left factor.

(e) **False.** $A = \operatorname{diag}(1, 0)$ and $C = \operatorname{diag}(0, 1)$ are nonzero but $AC = 0$.

(f) **False.** Expanding gives $A^2 - B^2 + BA - AB$. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \operatorname{diag}(1, 2)$, one has $AB \neq BA$, so the extra term does not vanish.

(g) **True.** A row swap or nonzero row scaling of I_n leaves exactly n nonzero entries. Adding a nonzero multiple of a different row introduces one off-diagonal nonzero entry, giving $n + 1$. A zero-multiple replacement leaves the identity with n entries.

(h) **True.** A swap matrix and a scaling matrix are symmetric. The transpose of $I + re_i e_j^T$ is $I + re_j e_i^T$, which is the matrix for the corresponding row replacement with the indices exchanged.

(i) **True.** An elementary matrix is obtained by one row operation on a square identity matrix; its shape remains square.

(j) **False.** Every product of elementary matrices is invertible, whereas a square zero matrix is not. Only invertible square matrices admit such a factorization.

(k) **True.** Three pivots in a 3×3 matrix imply reduced form I_3. List the elementary row operations that carry out that reduction; their matrices $E_1, \ldots, E_p$ give $E_p \cdots E_1 A = I$.

(l) **False without a square hypothesis.** Take $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $AB = I_2$ but A is rectangular and has a nonzero null vector e_3, so it has no two-sided inverse. If A, B were assumed square, the assertion would instead be true by the IMT.

(m) **False.** The inverse is $B^{-1}A^{-1}$, with reversed order. For the noncommuting invertible pair in part (f), the same-order product is not an inverse.

(n) **True.** From $AB = BA$, left-multiply by A^{-1} and then right-multiply by A^{-1}: $BA^{-1} = A^{-1}B$.

(o) **False.** The correct scalar factor is $1/r$: $(rA)((1/r)A^{-1}) = I$. Taking $A = I$, $r = 2$ disproves the claimed factor r.

(p) **True.** Let x_0 be the unique solution to the stated right side. If $Az = 0$, then $A(x_0 + z) = Ax_0$, so uniqueness forces $z = 0$. Thus the square matrix has trivial null space and is invertible.

Exercise 2.S.2: Invert a supplied inverse

Since $C^{-1} = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$ has determinant $28 - 30 = -2$, invert this 2×2 matrix:

$$\boxed{C = \frac{1}{-2}\begin{bmatrix} 7 & -5 \\ -6 & 4 \end{bmatrix} = \begin{bmatrix} -7/2 & 5/2 \\ 3 & -2 \end{bmatrix}}.$$

The inverse of an inverse is the original matrix, which explains why this operation recovers C rather than another new matrix.

Exercise 2.S.3: A nilpotent shift and a finite geometric sum

The given A sends $(x_1, x_2, x_3)^T$ to $(0, x_1, x_2)^T$. Therefore $A^2x = (0, 0, x_1)^T$ and $A^3x = 0$ for every x, proving $A^3 = 0$. Distribute the requested product in its existing order:

$$(I - A)(I + A + A^2) = I + A + A^2 - A - A^2 - A^3 = I.$$

The cancellation works because all terms are powers of the same matrix.

Exercise 2.S.4: Invert identity minus a nilpotent matrix

Let $S = I + A + \cdots + A^{n-1}$. Then telescoping gives

$$(I - A)S = I - A^n = I.$$

Multiplying in the other order gives the same telescoping sum, $S(I - A) = I - A^n = I$, since a matrix commutes with its own powers. Thus $\boxed{(I - A)^{-1} = I + A + \cdots + A^{n-1}}$. No inverse of A was used; in fact a nilpotent matrix of positive size cannot itself be invertible.

Exercise 2.S.5: Reduce higher powers using a quadratic identity

The hypothesis says $A^2 = 2A - I$. Multiply by A and substitute again:

$$A^3 = 2A^2 - A = 2(2A - I) - A = 3A - 2I.$$

Next multiply this result by A:

$$A^4 = 3A^2 - 2A = 3(2A - I) - 2A = 4A - 3I.$$

Every step uses multiplication and addition of already equal matrices; we did not divide by A or assume it was diagonalizable.

Exercise 2.S.6: A pair of anticommuting matrices

Direct multiplication gives

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -AB.$$

Thus both are their own inverses, but they anticommute rather than commute. This is a concrete illustration of behavior unavailable for nonzero real scalars, since scalar multiplication commutes.

Exercise 2.S.7: Solve a two-column matrix equation efficiently

Compute $A^{-1}B$ by solving $AX = B$. Begin with $[A \mid B]$. Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - (1/2)R_2$:

$$\left[\begin{array}{ccc|cc} 1 & 3 & 8 & -3 & 5 \\ 2 & 4 & 11 & 1 & 5 \\ 1 & 2 & 5 & 3 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|cc} 1 & 3 & 8 & -3 & 5 \\ 0 & -2 & -5 & 7 & -5 \\ 0 & 0 & -1/2 & 5/2 & 3/2 \end{array}\right].$$

Back-substitution gives the last row of X as $(-5, -3)$, then the second as $(9, 10)$, and the first as $(10, -1)$. Therefore

$$\boxed{A^{-1}B = \begin{bmatrix} 10 & -1 \\ 9 & 10 \\ -5 & -3 \end{bmatrix}}.$$

The three nonzero pivots verify invertibility of A, and exact multiplication checks $AX = B$. No entire inverse was computed.

Exercise 2.S.8: Determine a map from two prescribed images

Place the input vectors into $X = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and outputs into $Y = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$. The two prescribed equations are exactly $AX = Y$. Since $\det X = 7 - 6 = 1$, $X^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. Thus

$$\boxed{A = YX^{-1} = \begin{bmatrix} -2 & 1 \\ 4 & -1 \end{bmatrix}}.$$

Indeed $A(1, 3)^T = (1, 1)^T$ and $A(2, 7)^T = (3, 1)^T$. The independent inputs ensure that this matrix is uniquely determined.

Exercise 2.S.9: Recover the left factor of a known product

The given B has determinant $7-6=1$, so $B^{-1}=\begin{bmatrix}1 & -3\\ -2 & 7\end{bmatrix}$. Right multiplication by this inverse yields

$$A=(AB)B^{-1}=\begin{bmatrix}5 & 4\\ -2 & 3\end{bmatrix}\begin{bmatrix}1 & -3\\ -2 & 7\end{bmatrix}=\boxed{\begin{bmatrix}-3 & 13\\ -8 & 27\end{bmatrix}}.$$

For example the first output row checks as $[-3,13]B=[-21+26,-9+13]=[5,4]$.

Exercise 2.S.10: Another formula for an inverse

If A is invertible, so is A^T. Therefore A^TA is a product of invertible square matrices and is invertible. Set $X=(A^TA)^{-1}A^T$. Then $XA=(A^TA)^{-1}(A^TA)=I$. Since A^{-1} already exists, right multiplication by it gives $X=XAA^{-1}=A^{-1}$, establishing

$$\boxed{A^{-1}=(A^TA)^{-1}A^T}.$$

This is an algebraic identity, not a recommendation to compute the inverse by forming A^TA in floating-point arithmetic.

Exercise 2.S.11: Vandermonde matrices and interpolation

(a) Row i of Vc is $c_0+c_1x_i+\cdots+c_{n-1}x_i^{n-1}=p(x_i)$. Thus $Vc=y$ states precisely $p(x_i)=y_i$ for every i.

(b) Suppose $Vc=0$. Then p vanishes at the n distinct numbers $x_1,\ldots,x_n$. A nonzero polynomial of degree at most $n-1$ can have at most $n-1$ distinct roots, so p must be the zero polynomial. All its coefficients c_j are therefore zero. Thus $Vc=0$ has only the trivial solution, proving that the columns of V are independent.

(c) The square matrix V is invertible by the IMT, using part (b). For every arbitrary data vector y, choose $c=V^{-1}y$ and form the polynomial with those coefficients. Part (a) shows it interpolates every data point, and its degree is at most $n-1$. The polynomial is also unique: any second coefficient vector would be another solution of $Vc=y$.

Exercise 2.S.12: Read the first two columns of an LU product

The first column of upper triangular U is $(u_{11},0,\ldots,0)^T$, so $a_1=Lu_1=u_{11}l_1$. The second column of U is $(u_{12},u_{22},0,\ldots,0)^T$, so $a_2=u_{12}l_1+u_{22}l_2$. Thus the first column of A is a multiple of column 1 of L, and the second is a combination of columns 1 and 2 of L. Triangularity, not a mysterious cancellation, makes the later coefficients zero.

Exercise 2.S.13: Projection and reflection identities

Since $u^Tu=1$ is a scalar, associativity gives

$$\text{(a)}\quad P^2=(uu^T)(uu^T)=u(u^Tu)u^T=uu^T=P.$$

For the transpose,

$$\text{(b)} \quad P^T = (uu^T)^T = (u^T)^T u^T = uu^T = P.$$

Finally distribute the product using $P^2 = P$:

$$\text{(c)} \quad Q^2 = (I - 2P)^2 = I - 4P + 4P^2 = I.$$

Thus Q undoes itself: applying the same reflection twice returns every vector to its original position.

Exercise 2.S.14: A coordinate-plane reflection

With $u = e_3$,

$$P = uu^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad Q = I - 2P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

For $x = (1, 5, 3)^T$ this gives $Px = (0, 0, 3)^T$ and $Qx = (1, 5, -3)^T$. The vector $x - Px = (1, 5, 0)^T$ is the component in the coordinate plane; Q preserves it and reverses the perpendicular component Px.

Exercise 2.S.15: A product of elementary operations gives row equivalence

Multiplication by E_1 performs one elementary row operation on B, then E_2 performs another and E_3 a third. Consequently $C = E_3E_2E_1B$ is obtained from B by a finite sequence of row operations, which is the definition of row equivalence. The reverse sequence uses $E_3^{-1}, E_2^{-1}, E_1^{-1}$ in that temporal order, so the relationship works both ways.

Exercise 2.S.16: Construct a nonzero matrix with zero product

Because square A is singular, $Av = 0$ has a nonzero solution v. Choose one by row reduction, then set $B = [v \ 0 \ \cdots \ 0]$. The first column makes $B \neq 0$, whereas $AB = [Av \ A0 \ \cdots \ A0] = 0$. This construction works even if the null space is only one-dimensional.

Exercise 2.S.17: A product that factors through four dimensions

Every column of AB is a combination of the four columns of the 6×4 matrix A. Therefore $\text{Col}(AB) \subseteq \text{Col}\, A$, whose dimension is at most four. It cannot equal $\mathbb{R}^6$, whose dimension is six. The 6×6 matrix AB is thus not onto and cannot be invertible by the IMT.

Exercise 2.S.18: A left inverse gives uniqueness for attainable outputs

If $Ax = b$, then $x = CAx = Cb$. Thus every solution must equal Cb. The premise already says a solution exists, so it is unique. Equivalently, if $Ax_1 = Ax_2$, multiplication by C yields $x_1 = x_2$. The 5×3 matrix need not attain every $b \in \mathbb{R}^5$; the proof establishes uniqueness when existence has separately been supplied.

Exercise 2.S.19: Observe and explain limits of stochastic powers

Both matrices have nonnegative entries and each column sums to one. Such a matrix is called column-stochastic: it preserves the sum of a vector's coordinates. Exact rational powers give

$$A^2 = \begin{bmatrix} .31 & .26 & .30 \\ .39 & .48 & .39 \\ .30 & .26 & .31 \end{bmatrix}, \qquad B^2 = \begin{bmatrix} .29 & .18 & .18 \\ .33 & .44 & .33 \\ .38 & .38 & .49 \end{bmatrix}.$$

For $k = 8$, the powers are approximately

$$A^8 \approx \begin{bmatrix} .285728 & .285696 & .285728 \\ .428543 & .428609 & .428543 \\ .285728 & .285696 & .285728 \end{bmatrix}, \quad B^8 \approx \begin{bmatrix} .202364 & .202218 & .202218 \\ .370732 & .370879 & .370732 \\ .426904 & .426904 & .427050 \end{bmatrix}.$$

By $k = 16$ their columns are close to the respective vectors

$$p_A = (2/7, 3/7, 2/7)^T, \qquad p_B = (18/89, 33/89, 38/89)^T.$$

These are verified exactly by $Ap_A = p_A$, $Bp_B = p_B$, and coordinate sums one. The powers tend toward matrices having every column equal to the corresponding p. An optional explanation from later eigenvalue theory is that A's other eigenvalues are $.3, .1$, and B's are $\pm\sqrt{11}/10$; all have magnitude less than one, so their powers die out.

Here is a further reproducible investigation with positive column-stochastic matrices:

```
import numpy as np
rng = np.random.default_rng(22019)
C = rng.integers(1,10,(3,3)).astype(float)
C = C/C.sum(axis=0)
print(np.linalg.matrix_power(C,16))
```

The executed result has nearly identical columns $(.20882917, .35930902, .43186180)^T$. A precise conjecture supported by these examples is that positive stochastic matrices, and more generally stochastic matrices with some positive power, have powers converging to a matrix with identical columns. It would be false to remove that extra condition: the permutation matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is stochastic but its powers alternate instead of converging. Thus the experiment suggests a qualified theorem, not a claim about every stochastic matrix.

Exercise 2.S.20: A closed formula for inverses of zero-diagonal ones matrices

Let J_n have every entry equal to one. Then $A_n = J_n - I_n$ and $J_n^2 = nJ_n$, since each entry of the square is a sum of n ones. For $n \geq 2$ propose

$$B_n = -I_n + \frac{1}{n-1} J_n.$$

Its diagonal entries are $-(n-2)/(n-1)$ and off-diagonal entries are $1/(n-1)$. In particular the computed inverses for the three requested sizes are

$$B_4 = \frac{1}{3} \begin{bmatrix} -2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 1 \\ 1 & 1 & -2 & 1 \\ 1 & 1 & 1 & -2 \end{bmatrix},$$

$$B_5 = \frac{1}{4}\begin{bmatrix} -3 & 1 & 1 & 1 & 1 \\ 1 & -3 & 1 & 1 & 1 \\ 1 & 1 & -3 & 1 & 1 \\ 1 & 1 & 1 & -3 & 1 \\ 1 & 1 & 1 & 1 & -3 \end{bmatrix},$$

$$B_6 = \frac{1}{5}\begin{bmatrix} -4 & 1 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & 1 & -4 \end{bmatrix}.$$

To turn the conjecture into a proof, multiply symbolically:

$$(J_n - I_n)B_n = I_n + \left(-1 + \frac{n}{n-1} - \frac{1}{n-1}\right) J_n = I_n.$$

The reverse product is identical because both factors are combinations of I_n and J_n. Hence $B_n = A_n^{-1}$ for all $n \geq 2$. For $n = 1$, $A_1 = [0]$ is singular and the denominator $n-1$ correctly signals the excluded case.